

A Safe Approach to Defeating Malicious Drones in a Civilian Environment

Benjamin Ashurov 20093355

Mechanical Engineering Department, University College London

ABSTRACT

The threat posed by the misuse of drones is a clear and present danger to the public, with no effective countermeasures. This report investigates various aspects of this problem and proposes an effective approach to defeating malicious drones. By analysing the strengths and weaknesses of existing technologies, it was determined that an ideal solution must reliably defeat publicly available drones while minimising the chance of collateral damage. To find this solution, a first principles-based brainstorming approach was employed to generate original ideas, and a quantitative comparison table was used to assess their merits objectively. After careful refinement, a novel concept was discovered: a net-based rocket interceptor. This solution postulates that the best way of defeating drones is to adapt established anti-aircraft rocket systems to deploy a net that captures a targeted drone. Consequently, the interceptor will be able to transport the target to a safe location, thereby removing the threat from the defended area in a controlled, explosion-free manner. To design the various technical characteristics of this system, a comprehensive list of requirements and criteria was formulated based on existing rockets. This research informed the overall dimensions of the system, as well as the targeting and propulsion mechanisms chosen for the interceptor. Furthermore, it was found that the capture mechanism is a completely novel concept that cannot be based on similar technology and must be designed from the ground up. This was achieved by performing rigorous material selection, FEA analysis of load-bearing components and detailed CAD modelling of the deployment mechanism. As a result of this work, carefully selected rocket technology has been combined with a highly detailed capture mechanism design, forming a highly optimised solution for this problem. This was done while prioritising reliability, simplicity, and low manufacturing cost throughout the design. Consequently, the final product is not only effective, but also accessible and can be deployed on a wide scale by government agencies, allowing them to largely neutralise this threat. Therefore, an effective approach to dealing with malicious drones in a civilian environment has been achieved.

Keywords: Interceptor, Design, Rocket, Capture Mechanism

1. PROJECT BACKGROUND

Quadcopter drones are a type of unmanned aerial vehicle (UAV) that use 4 propellers to achieve stable hovering and flight. In the last ten years, small commercial quadcopters have seen significant improvement in their quality, availability and ease of use. In 2020, over 5 million units were sold [1], with DJI Tech leading the market with 76% of all sales. Their flagship product is the Mavic 3, a “camera” drone, which combines data from gyroscope, accelerometer and ultrasonic distance sensors as an input for a sophisticated control algorithms that produces stable flight. This allows for simple control by the operator, who simply provides movement directions based on the drone’s image output.

The other common type of quadcopter is known as the first person view (FPV) drone, which is categorised by a lack of control stabilisation algorithms. Instead, these drones have full freedom of movement with the operator controlling the pitch, yaw and roll independently, allowing for highly complex maneuvers and fast flight. Conversely, this makes flying more difficult, which is why VR goggles are typically used to broadcast the camera

feed, thereby increasing the pilot’s awareness and giving this drone type its name.

Commercial drones have proven themselves to be a powerful technology, which is now utilised throughout the agricultural sector as well as in law enforcement. Conversely, they have also been used in a variety of criminal activities, including drugs delivery, interference with aircraft and espionage. A prominent example of drone misuse was the shutdown of Gatwick airport in 2018 for 36 hours due to unauthorised drone flights [2]. However, the greatest danger from drones arises from their use for targeting civilian objects with explosives. This threat has been showcased throughout the war in Ukraine where camera drones have been used to drop munitions and FPV drones have had explosives attached to them and used as suicide drones by purposely flying into the targets.

2. PROBLEM IDENTIFICATION

The accessible nature of this technology means there is a clear and present danger of their use in a terrorist attack. Groups such as Al Shabab and Hamas have employed drones during combat [3], and it is easy to

imagine how a terrorist with access to a small amount of explosives can use a drone attack an airport or a public gathering.

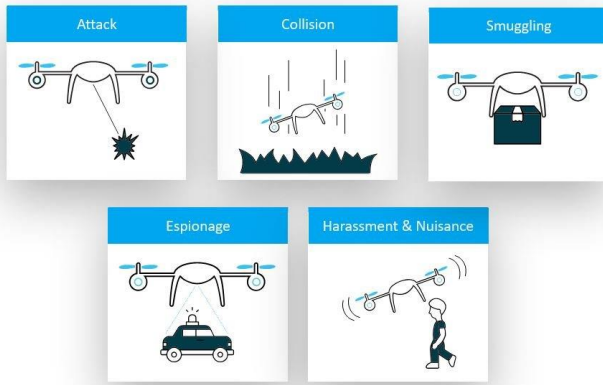


Figure 1 - Types of Malicious Drone Activities[3]

Several technologies have been developed to help address this threat. The most widespread strategies today are variations of the “soft” approach, which involve interfering with a drone’s command signals. These approaches are relatively cheap, simple and somewhat effective against commercial drones; however, their effect can be significantly reduced through simple modifications. For example, jamming, is the most widely used technique today and involves broadcasting a signal at the same frequency as the drone. However, it can be countered by operating the drone at a different frequency to the jammer. In addition, advancements in AI are predicted to make drones completely self-sufficient, nullifying the soft kill approach entirely as the signal from an operator is no longer required.

Another method, which is primarily used to defend against drones in the military context is known as the “hard” approach. This involves using a physical impact, such as anti-aircraft rockets that explode next to the target drone to incapacitate it. However, this approach is inherently unsuitable for use in a civilian environment, as explosives produce fragmentation, which are likely to result in collateral damage. In addition, many anti-aircraft rockets currently in use are somewhat ineffective against drones, as they are designed to target aircraft which are much larger.

Furthermore, novel approaches have been developed to disable malicious drones, including lasers and microwave-based systems [4]. These systems are still not entirely suitable, as they cause drones to drop out of the sky in an uncontrolled manner. This is inherently dangerous as a drone packed with

explosives falling in a civilian area presents an unacceptable amount of risk of collateral damage.

PROJECT AIM

Therefore, this project aims to develop a method that has a high probability of neutralising rogue drones while minimising the chance of collateral damage.

3. CONCEPTS EXPLORATION

A number of techniques, including rapid ideation, analogous brainstorming and TRIZ analysis were used to generate five innovative concepts to approach this challenge:

3.1. Brainstorming

1. Hail Generator- Small blocks of ice are fired at the targeted drone, imitating hail which is effective at preventing drone flight. Despite the reduced danger compared to explosive carrying missiles, drones will still be falling from the sky in an uncontrolled manner, which is dangerous.

2. Sensor Signal Generator- Drone elevation sensors and ultrasonic emitters are interfered with by emitting an electric and ultrasonic signal that suggest that its elevation is much higher than instructed, causing the target to automatically land. This will not be effective against user piloted drones, who are likely to override the landing instruction, subverting the attack.

3. Physical barrier generation- External command signals can be blocked by scattering reflective materials in the air between the target and the operator. This approach, which is inspired by chafe, can be effective, but requires detailed experimentation with materials, as no known material is obviously effective at deflecting radio signals.

4. FPV Drone ramming- A fast, manually controlled drone can be equipped with a ram or a tesla coil which would knock the targeted drone out of the sky. Even though drone ramming is a proven approach, the target will fall from the sky, which is not safe.

5. Net shooting drones- A large, fast drone carries a net gun within range of the target. When in range, the targeted drone is captured and transferred to a safe location. This approach is limited by the requirement for a skilled operator who can only target one drone at a time and the difficulty of catching a similar speed target [5].

To decide which one of these concepts is the best to solve this problem an AHP [6] table was constructed with five equally weighted judging criteria. Solution requirements include a high success probability, dubbing it “effective”, as well as being creative since

none of the currently deployed conventional methods have been proven to work. In addition, the solution must also be feasible to design with limited resources of a undergraduate project, practically usable in many environments and low in cost.

	Hail generator	Sensor Signal Interference	Physical Barrier Generation	FPV Drone Ramming	Net shooting drone
Feasibility	3	4	4	4	5
Practicality	3	5	4	4	4
Price	5	4	3	5	5
Effectiveness	3	5	4	4	3
Creativity	5	4	4	3	2
Score	19	22	19	20	24

Table 1- AHP table comparison of the various anti- drone concepts

3.2 Concept Selection and Refinement: After objectively analysing the pros and cons of these concepts, it was determined net shooting capture drone was the most promising approach to this problem. To improve this solution, its primary weaknesses must be addressed, particularly by increasing automation and reducing the possibility of the target evading the capture drone. With the help of the TRIZ matrix, it was determined that a more conventional method of “delivering” the net to the drone should be attempted, such as a rocket, instead of a drone. This method would have the advantage of being much faster than a drone, reducing the chance of evasion. In addition, modern anti-aircraft rockets also require little to no user input to track and hit their targets. Therefore, all the target drones can be engaged simultaneously, and many skilled pilots would not be needed.

Instead of a conventional explosive warhead, this rocket would have an inbuilt net that would expand from its sides and capture the drone, as showcased in Figure 2. After capture, the net and the target will both be retracted into the rocket, which would proceed to land safely in a preprogrammed location. This approach promised to allow for a reusable device that will capture the targeted drone in a reliable manner without knocking the drone out of the sky, which may cause damage when falling.

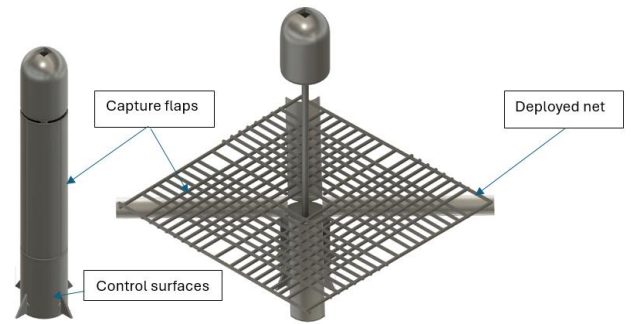


Figure 2 - CAD of the Interceptor Concept in the Closed (left) and Open (right) Positions

3.3. Design Criteria

To inform the design decisions for the interceptor, comprehensive design requirements and criteria must be formulated. These are primarily based on the range of characteristics of other rockets used for intercepting airborne objects.

Specifically, the short-range air defense (SHORAD) category of rockets was selected as the basis for the design, due to a similar use profile. Furthermore, the drone that the system would be designed to capture is also considered. Specifically, the DJI Mavic 3 drone was selected as the basis for the design as the most widely used drone in the world [2].

Feature	Relevance	Quantitative Range	Justification
Length	Minimised to increase ease of use	0.5m to 2.5m	Shortest [9] and longest [10] existing short range air defence rockets.
Width	Must be able to fit the captured drone inside.	85mm to 533mm	The smallest dimension on a drone [11] and widest air defence rocket [12].
Velocity	Must catch up with the drone and defeat evasion manoeuvres	21ms-1 to 110ms-1	Max speed of the drone and of a comparable rocket
Weight	Smaller weight increases manoeuvrability and ease of use	4kg to 30kg	Lightest [9] and heaviest [10] existing short range air defence rockets.
Toughness	Absorb the impulse from the collision with a drone	1.46 KJ	Based on the collision velocity and the mass of the drone (0.734kg)
Explosive resistance	Continue flight if explosive carried by a drone detonates	Up to 0.785MJ	Energy released by 0.5 kg (drone capacity) of C4
Safety	Contingencies to prevent collisions with non-targets	Compliance with 14 CFR Chapter III	Regulations governing High Power Rocket Motors near airports [13]
Deployed net area	Larger net increases the probability of drone capture	$3.14m^2 - 19.6 m^2$	Based on the lengths of the capture mechanism
Price	Determines the commercial viability of the product	£500-£1200	Estimated materials and manufacturing cost
Reusability	Increases the sustainability of the design and reduces price per use	10 Uses	Typical number of times a reusable rocket can be used [14]
Reliability	Increases the likelihood of a successful intercept	Probability of failure <0.1%	Standard for uncrewed rocket [15]

Table 2 - Interceptor Design Requirements and Criteria

4. INTERCEPTOR DESIGN

To develop the theoretical interceptor concept into a practically viable solution, its component dimensions, guidance, propulsion, and capture mechanisms must be designed. The dimensions, specifically the height and width of the interceptor, are based on the targeted drone's dimensions as well as the precision level of the rocket. Additionally, the guidance and propulsion systems are adapted from existing technology, particularly the Robot System 70 Bolide anti-aircraft rocket. Finally, after the capture mechanism was described in general terms, after which it was found that more detailed technical design is required, as seen in section 5.

4.1 Interceptor Dimensions

The diameter of the system should be as small as possible to decrease the system's mass and, therefore, the size of the motor. However, the rocket must be able to contain the Mavic 3 (96×93×221 mm) drone inside after capture. Assuming the drone is lying flat in the net when captured, to accommodate it in the space between the internal support rod and the wall, the a

space of 93 mm is required. Accounting for the size of the internal rod with a diameter of 40 mm and a small offset of 3 mm, the diameter of the capture mechanism is 221 mm.

Similarly, the system height should be minimised to ease usage in the field, as well as to decrease the mass. Conversely, a greater height will allow for longer capture flaps, which can accommodate a larger net and decrease the interceptor's precision requirements. In a classic anti-aircraft rocket like the Robot Systems 70 BOLIDE, a proximity fuse is fitted, causing it to detonate if it passes within a preset distance from the target. Similarly, by increasing the size of the net, the rocket can somewhat miss the targeted drone but still capture it. By investigating the BOLIDE rocket, which carries 1.1 kg of explosives, it was found that it will be triggered to explode by the proximity fuse when passing 2.6 m away from the target [16]. This is calculated by assuming the BOLIDE rocket has the same overpressure requirements produced by the detonation to damage the target as the STRELA 10, where:

$$\text{Overpressure} = K * \frac{(\text{Explosive Weight})^{\frac{1}{3}}}{\text{Distance}} \quad \text{Eq 1}$$

Since its proximity fuse distance (4 m) and explosive weight (3 kg) are publicly available, the corresponding distance of 2.6 m for the BOLIDE ROCKET was found. By assuming the same level of precision for the interceptor, this distance can be taken as the equivalent to the required length of the capture flaps. Therefore, when accounting for the propulsion and guidance mechanism, the overall system length is 3.2 meters.

4.1 Guidance Mechanism

The guidance element of a rocket is designed to spot the target and provide information to the rocket's propulsion and steering units to move in a way that ensures the drone is hit. Most common types of guidance include radar-based systems, which emit radio waves and record their reflection, as well as thermal targeting systems that lock onto a source of heat and guide the rocket toward it. However, both methods have issues with spotting drones due to their small dimensions and low heat signatures. Another solution, which is utilised by the BOLIDE rocket [7], is to use a visual feed to spot the drone and shine a laser beam at it, which the rocket can follow to its target. This guidance system type was chosen for the interceptor due to its proven effectiveness against drones [8]. Furthermore, it has a relatively low price point compared to radar or heat-based seekers, as only simple laser receivers placed at the tips of the fins are needed for it to work.

4.2 Propulsion Mechanism

Similarly, the propulsion of the rocket can also be based on the BOLIDE rocket, using a solid fuel rocket motor and four movable tail fins. To allow for reusability, the solid fuel motor must be modular, making it easy to replace, and a deployable parachute must be added to the aft of the rocket to allow for a smooth descent at a predetermined location when the drone is captured. Furthermore, the power output, and therefore the speed of the interceptor must be much lower than that of the BOLIDE rocket, which flies at 686 m/s. This is based on the fact that drones move much slower than the maximum speed of the SU-27 [8] (389 m/s) which the BOLIDE rocket is optimised to intercept. In addition, a collision at very high speeds can cause the drone to shatter and the explosive it is carrying to detonate.

To approximate the ideal speed that would allow the rocket to intercept its target, the ratio between the maximum speed of the BOLIDE rocket and its intended target, the SU-27, was calculated to be 1.76. The same ratio can then be applied to this interceptor design, where the maximum speed of the Mavic 3 drone is 21 m/s, so the target speed of the interceptor was determined to be 37 m/s. This lower speed also helps adapt the interceptor for safe use in a civilian setting by reducing the scale of potential damage if it hits anything other than its intended target.

4.3 Capture Mechanism

Finally, the capture mechanism of the rocket will consist of “capture flaps” that fold out from the rocket's base and suspend a net between their ends, as seen in Figure 2. This process will occur as the rocket approaches the targeted drone, which will be caught in the net when the rocket passes within proximity of the target. Note that a direct impact is not preferable, and the system must ensure an offset of its centre tip from the drone. After interception, the capture flaps will fold back in, and the drone will be held inside the rocket's body. This mechanism is broadly similar to the setup used in the Hellfire R9X [29], which also deploys extendable elements from its middle section before impact.

While considering the number of capture flaps, it was noted that a smaller number would decrease the complexity, and therefore the cost and failure rate. Conversely, a minimum of three flaps is necessary to suspend the net in 3D space, and more flaps increase the net's area and therefore probability that the drone will be caught with a system of a given accuracy. To balance these considerations, the four-flap design was chosen, which will form a square-shaped net that will cover 64% of the possible area if the net were a perfect circle. Despite this, the system remains relatively simple with only one more flap than absolutely necessary.

Based on the design work done in this section, it was determined that further technical analysis of the capture mechanism must be carried out, as it can't be based on existing technology.

5. CAPTURE MECHANISM DESIGN

Three principal component groups must be designed to form a working capture mechanism. These include the deployment mechanism, which will move the

capture flaps from the closed to the open position, as well as the load-bearing components, which support the net and absorb the impact with the drone. In addition, the net itself must be selected and designed to survive violent impacts and to fold back into the interceptor when the capture flaps retract.

5.1 MATERIAL SELECTION

To select the best materials for the deployment and load-bearing mechanisms in a systematic manner, a strategy was devised using the Edupack Level 3 library. First, general constraints were applied for both component groups to filter among the 3,900 materials in this library, including limiting material pricing below £20/kg to ensure the interceptor is economically viable to manufacture. Also, the materials must be non-flammable and resistant to freshwater to retain physical integrity in case of onboard fire and to allow for the interceptor to be used in the rain. In addition, to fulfil the criteria of sustainable design, the considered materials were filtered to be somewhat recyclable and minimise energy usage during processing. Next, the desirable material properties were selected for each component type based on their design criteria. These were then used to rank the various possible materials and select the best one for that particular application.

The load-bearing components include the base, the capture flaps, the hinge, and the rotating rods, as seen in Figure 10. Before the materials were plotted based on their desirable characteristics, two extra constraints were added. First, high thermal resistance, which would ensure that the rocket maintains its integrity in case the targeted drone carries explosives that detonate. Second, the material must have a high yield strength, which is set at above 1.5 GPa, as plastic deformation should be minimised during use. Next, the desirable parameters were determined, with the primary one being fracture toughness, to reduce the chance that manufacturing defects will propagate through the component, causing failure after impact. The other key requirement for these components is high stiffness to reduce elastic deformation that could result in collisions with other components or place undue strain on the net. When the materials that passed the above-mentioned constraints were plotted against these criteria, as seen in Figure 3, the steel alloy “Fe-5Cr-Mo-V aircraft steel” showcased the best overall performance. Furthermore, it has a

relatively low price of £17.2/kg, which is desirable, and is currently used in load-bearing applications for aircraft components, such as landing gear, therefore demonstrates the soundness of this choice.

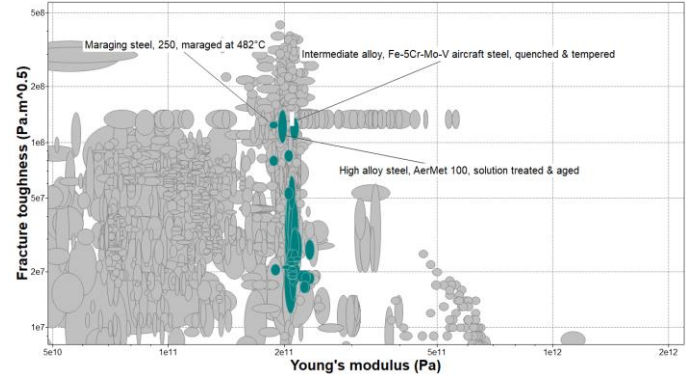


Figure 3 - Potential Load Bearing Materials

Next, the material used for the deployment mechanism, including the gears, the gear shafts and the motor mount were derived. The design of these complex components should be cost conscious as complex features of these components, such as the gear teeth, can significantly drive up the manufacturing time, and therefore pricing of the product [18]. Their design necessitates the use of machining due to its high precision requirements. Therefore, one of the optimised parameters was selected to be machining speed, which should be maximised. Another parameter to optimise for is the overall mass of the component, which contributes to the required size of the booster stage needed to provide necessary acceleration for the rocket and also drives cost. To this end, the other considered parameter is the material density, which should be minimised. The resultant plot helped identify Magnesium ASTM 9980A as the best option for the deployment mechanism, which was corroborated by its use in aerospace applications.

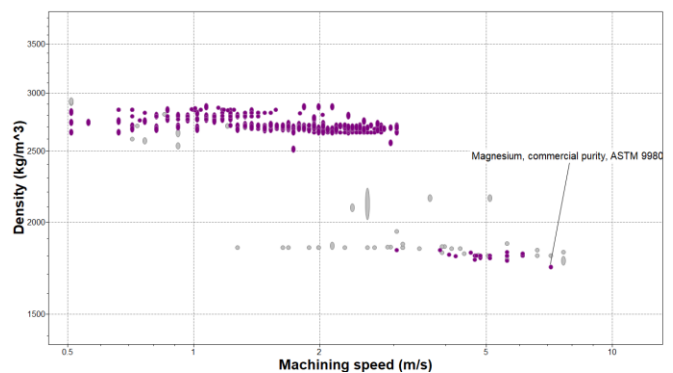


Figure 4 - Potential Deployment Mechanism Materials

5.2 DEPLOYMENT MECHANISM

To allow the capture flaps to move between the closed and open positions, they must be suspended on hinges and actuated by a force-generating mechanism. Various approaches to this problem were considered, including a hydraulic-based design used to drive wing flaps on aeroplanes, offering high operational speeds and torque. However, the major downsides of this approach include complex maintenance involving hydraulic fluids, as well as the high price and weight, making it untenable. Another considered method was to use bleed-off gases to drive the capture flaps, similar to the way they are used to move control surfaces on the Stinger rocket. The problem with this approach would be the complex modifications required to the existing rocket design to channel the gases, which is undesirable. In the end, an electric motor-based approach was selected due to its simplicity and ability to generate high torque and speed. Several electric motor configurations were explored for this use case, including four separate motors, each driving a capture flap, and a larger, central motor connected to a gear that drives all four capture flaps. The latter design was chosen because high torque is needed to close the capture flaps due to the high air resistance forces acting on them. This mechanical system will allow the torque from a motor to be increased by an order of magnitude while reducing the total mass and cost of the system. The transmission quasi-direct drive motor made by ASF Tech [19] was chosen in this design, as it can generate up to 120 Nm of torque and has an acceptable weight (1.4 kg), RPM (100) and price (£192) for this application.

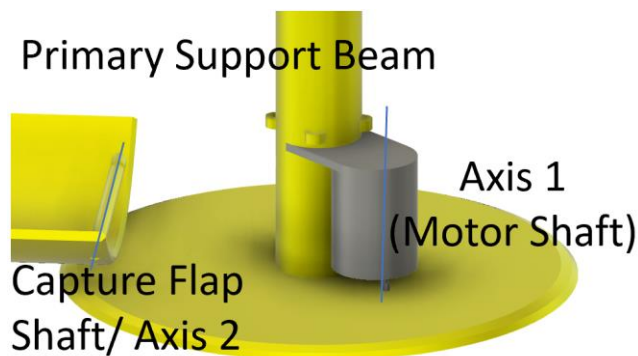


Figure 5 - Capture flap and Motor Axis

The initial CAD of this concept, as showcased in Figure 5, placed the motor next to the primary support beam and the capture flaps at the edges of the

base. Since axes 1 and 2 do not intersect, an intermediate shaft, whose axis intersects both, must be incorporated into the design to mechanically connect the capture flap shafts to the electric motor. To achieve this, a sophisticated bevel gear mechanism was designed, as seen below.

A high-torque motor is placed adjacent to the primary support beam, with its motor shaft facing downwards. When powered on, this transfers motion to the internal gearing of gear 1. This gear, which is placed inside an indent to guide its motion, has a large diameter, allowing it to transfer impulse to all four of the capture flap mechanisms.

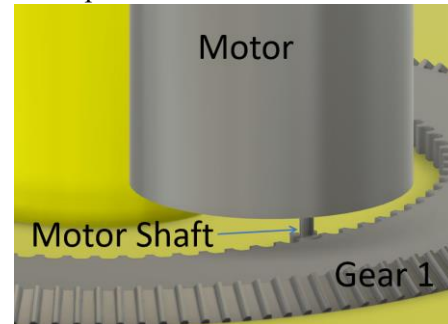


Figure 6- Motor Shaft and Gear 1 Pairing

The outside surface of Gear 1 is bevelled, allowing it to transfer motion to Gear 2 (Figure 7). These gears are positioned at 90 degrees to each other, so the motion is transferred to a plane perpendicular to the motor. Gear 2 is mounted on the intermediate shaft, whose axis intersects both axes 1 and 2.

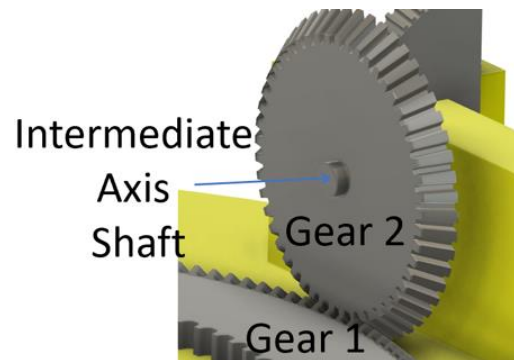


Figure 7 - Gear 1 and Gear 2 Pairing

A support block is added to the system to house the intermediate shaft and the capture flap shaft. These shafts are supported by embedded ball bearings, which are not modelled for simplicity, allowing them to rotate freely. The same shaft is also attached to Gear 3, which is bevelled in the opposite direction to Gear 2 and rotates together with it.

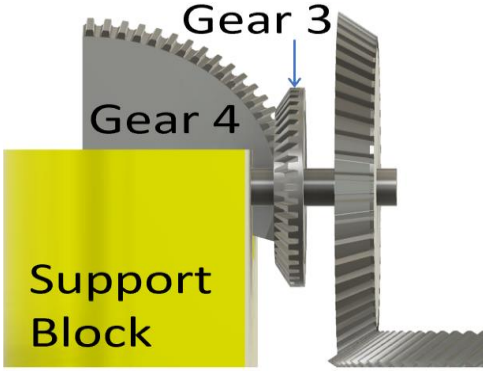


Figure 8 - Gear 3 and Gear 4 Pairing

The teeth of Gear 3 interlock with those of Gear 4, which is attached to the same shaft as the capture flap, causing it to rotate. Due to constrained dimensions, Gear 4 was designed as a partial gear with a 110-degree arc. This is sufficient because the capture flap itself only needs to move 90 degrees to transition between the open and closed positions. This gearing arrangement is replicated on all four sides, allowing a single motor and Gear 1 to drive all the capture flaps simultaneously. Specifically, the motor rotates counterclockwise, as seen from above, the capture flaps will swing open. This arrangement also ensures perfect synchronisation, as all the capture flaps are driven by one gear, preventing misalignment during opening or closing.

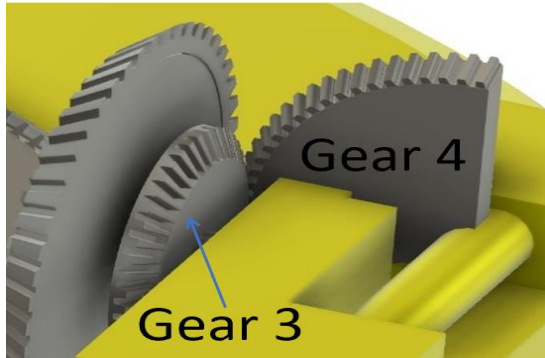


Figure 9 - Gear 3 and Gear 4 Pairing

The above mechanism relies on interlocking bevelled gear teeth to transfer motion from one gear to another. Since all gears vary in diameter, to ensure a tight fit of these teeth, their pitch cone angles must be complementary. For a given set of gears, the complementary pitch cone angle is described by the following functions:

$$\theta_1 = \arctan\left(\frac{D_1}{D_2}\right) \text{ Eq 2}, \theta_2 = \arctan\left(\frac{D_2}{D_1}\right) \text{ Eq 3}$$

This condition helps ensure that the gears' axes intersect at 90 degrees, and the angles of each gear's teeth are described in the table. Additionally, the gear ratio is calculated relative to the motor shaft, which has a diameter of 2 mm.

Gear number	Diameter (mm)	Teeth Angle	Gear ratio relative to the motor shaft
1	134	40.5	26.8
2	40	49.5	20
3	20.5	27.0	10.5
4	40	63.0	20

Table 3- Gear parameters

Based on the gear ratios with the motor, the rotational speed [20] and the torque [20] of the capture flaps shaft are calculated as follows:

$$\omega_{\text{final}} = \omega_{\text{initial}} * \frac{1}{\text{Gear ratio}} \text{ Eq 4}$$

$$\tau_{\text{final}} = \omega_{\text{initial}} * \text{Gear ratio} \text{ Eq 5}$$

Based on the parameters of the chosen motors, the speed of rotation of the capture flap is 5 RPM, so it will take 3 seconds to rotate through 90 degrees to fully open or close. Furthermore, since the total torque of the motor is distributed across four capture flaps, each shaft will rotate with a torque of 600 Nm. This is more than sufficient to overcome the 550 Nm of torque on the capture flap due to air resistance, as calculated in Section 5.3. Therefore, when the motor gear rotates clockwise, when viewed from above, the capture flaps will close.

All the gears use the same patterned tooth design, based on the default bevel gear tooth of this order of magnitude in size for Fusion 360. This tooth has a height of 1 mm and a width that varies between 0.6 mm and 0.5 mm along its 3 mm length. To transfer this design onto the gears, two scaled images of the cross-section of this tooth were placed on the surface of each gear and used to sketch an outline, which was then combined into a solid body using the loft command.

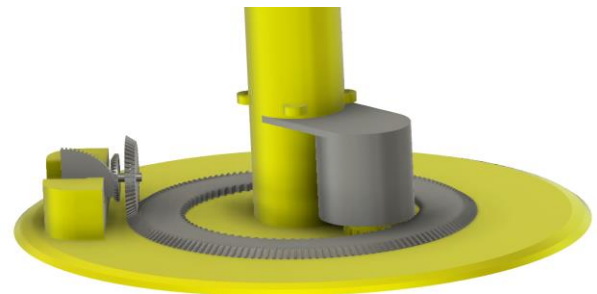


Figure 10 - Entire Deployment Mechanism

5.3 Load Bearing Elements

The load-bearing components of the interceptor are designed to absorb the impact with the drone without excessive deformation. Conversely, their design must accommodate the overall goal of minimising the total mass of the mechanism. Therefore, the thickness of the load-bearing elements, such as the support block (Figure 8), must be carefully selected. The initial dimensions of the components were modelled based on a field hockey goal, which has comparable strength requirements. The hooks that support the goals net must absorb the energy of an inelastic ball weighing 160 g and moving at speeds up to 44 m/s, which equates to 150 joules [21]. Since this is broadly the same order of magnitude of energy as that of a drone collision with the interceptor, the support hooks thickness of 2 mm was taken as the base minimum for the various load-bearing components [22].

To test and refine the validity of the initial design dimensions, the various applied forces were calculated in detail. Next, these were used to create a static force FEA model to simulate the maximum force applied on the net due to a collision with a drone, as well as the pressure on the deployed capture flap due to air resistance. The latter is expressed by the equation [23]:

$$P_{drag} = \frac{1}{2} * p * v^2 * C_p \quad Eq 6$$

Where p is the density of air (1.29 kgm^{-3}) [24], v is the velocity of the rocket (37 ms^{-1}), and C_d is the drag coefficient, which is taken to be equal to that of a rectangle flat plate (1.28) since that's the shape the capture flap faces the airflow. The total drag force was calculated to be 1.1 kPa.

The force due to the impact of the drone on the net-supporting components is also calculated. This is based on the stopping distance equation [25], assuming constant deceleration:

$$F_{Impact} = \frac{1}{2} * \frac{mv^2}{d} \quad Eq 7$$

Where m is the mass of the drone (0.89 kg) and v is the relative velocity between the rocket and the drone, which is the sum of their maximum velocities (58 ms^{-1}). Additionally, d is the impact distance (57 mm), which is equal to the height by which the net is suspended above the capture flap (see Figure 14). This was based on the assumption that all the KE will

be absorbed by the net, and the total force from impact was found to be 26.3kN.

Both the air resistance-induced pressure and the impact force, which are applied to the net-supporting components, were then applied to the system in an FEA model:

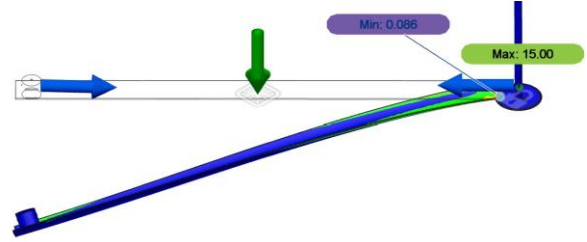


Figure 11 - Displacement of the Capture Flap

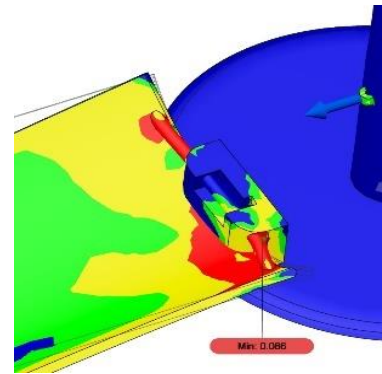


Figure 12 - Critical Deformation of Components

The initial dimensions of the design were found to result in unacceptable levels of deformation under applied loads. The tip of the capture flap was displaced by 32 mm, and a critically low safety factor was found in most of the flap shaft and a portion of the flap itself.

Based on these observations, the design was iteratively refined, with a repeated 1 mm step increase in the dimensions of the various components. First, the diameter of the shaft was increased in the areas that were shown to fail, from 7 mm to 12 mm, until the safety factor reached 1 throughout its length. Next, the thickness of the capture flap was increased from 3 mm to 5 mm to reduce the displacement distance. The final design, as seen in Figure 13, showcased a very small maximum deformation of 3 mm and a safety factor above 1 in all but three small areas of the support rod. This was deemed acceptable, as the considered forces represent the worst-case scenario—a head-on collision at the top speed of both the rocket and the drone. Additionally, further

increasing the thicknesses of the various components would also increase the mass, which must be minimised.

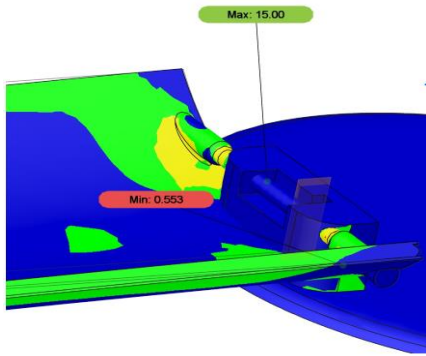


Figure 13 - FEA Results of the Revised Design

5.4 Net Composition and Deployment

The net used in this application must fulfil several design criteria, the pivotal ones being high mechanical strength and durability so that it retains structural stability during impact. Another requirement is flexibility, to ensure a large stopping distance of a captured drone, thereby reducing the force applied to the mounting components.

High-strength net manufacturing is an established field that involves complex design and quality control. Therefore, instead of performing a wholly original net design process, a selection among existing high-strength models was carried out. These included polyethylene hockey goal nets, Kevlar-based cargo nets used to secure objects during haulage, and a Dyneema polyethylene net used for industrial fishing. All of them have been shown to arrest falls of objects weighing over 40 kg, broadly qualifying them as “high strength” [26]. To select the optimal model, secondary characteristics such as mesh size and the presence of knots in the design were considered. These parameters are essential to ensure the drone remains intact for post-capture forensic analysis. Based on these considerations, the knotless, fine mesh (8 mm) Dyneema SK75-based net was selected.

This net model has a proven record of success in extreme stress environments. Specifically, it was used to catch Luke Aikins [27], who performed a dive from 7,600 m without a parachute. The net was used to arrest the fall of a 80 kg body travelling at a speed of 54 m/s, with a KE of 116 kJ. This is two orders of magnitude above the 1.5 kJ maximum KE of the targeted drone relative to the interceptor as calculated

in part 5.4. Therefore, this net type is guaranteed to survive the collision.

To quantitatively verify its suitability, FEA was carried out on a single strand of Dyneema SK75 fibre. The net, which has an area of 13.7 m², a mesh size of 8 mm and four fibres per mesh, consists of 855,625 fibres [28]. By assuming that the impact impulse is distributed evenly, it was found that the force applied to each fibre will be equal to 0.023 N. The FEA model verified that a fibre under these conditions will retain structural integrity, with very high a minimum safety factor of 7.8.

Finally, the net must be attached to the interceptor in such a manner that when the capture flaps swing out, as described in Section 5.2, it is positioned to catch the targeted drone. So, the net will be connected to the suspension points at the end of the capture flaps and to a support ring at the central column (Figure 14). When the target drone is captured, the net and the drone must retract back into the pre-deployment position. To facilitate this, a low-tension rubber band, with a spring constant of $k = 10 \text{ Nm}^{-1}$, is embedded in the net at a distance of 2.5 m from the centre. This helps ensure that the net sections between the capture flaps do not stick out when the interceptor closes.

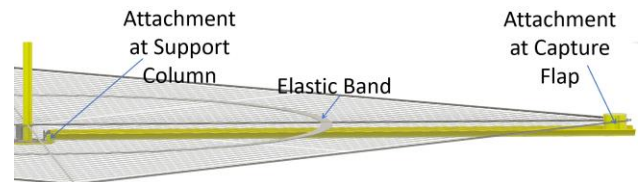


Figure 14 - Displacement (top) and Critical Deformation (bottom) of the Capture Flap

6. CRITICAL ANALYSIS

6.1 Work Outcomes

The initial outcome of this project was the development of a highly effective conceptual method for defeating drones in civilian environments. To achieve this, the strengths and weaknesses of existing approaches were identified and used to define the various aspects that needed to be addressed in an ideal solution. This analysis informed the idea generation stage, where five novel concepts were formulated, analysed, and refined. Overall, this method was very effective, resulting in an elegant, simple solution that can target any small drone type

and performs a controlled capture process which minimises the likelihood of collateral damage.

However, it was noted that one of the key elements of this solution—the use of rocketry—introduces safety risks and the potential for product misuse. This raises important legal and ethical concerns that have not yet been fully resolved. Additionally, some modern drones have been shown to fly faster than the 21 m/s maximum speed of the DJI Mavic 3 that the interceptor was based on. While these speeds are easily accessible by rockets, the design may need to be reassessed to create a universally applicable solution.

Next, this project outlined a detailed technical design process that demonstrated how this theoretical concept can be practically implemented. As a result of this work, the various components of the interceptor can be assembled to operate in the following, highly effective, interception sequence. Once a malicious drone is detected targeting technology, based on the RobotSystem 70 system, employs a laser beam riding system to guide the interceptor towards the target. This approach ensures a high degree of autonomy, simplicity, and accuracy within the system. The interceptor is propelled by a solid-fuel rocket at a high speed of 37 m/s, minimising the likelihood of successful evasion. Next, the guidance mechanism instructs the interceptor to deploy the capture flaps 120 metres away from the target. The capture flaps, driven by a high-torque motor and bevel gear system, swing out to stretch the net taut and form a rectangular shape (Figure 15). The targeted drone is ensnared in the net, which, along with the load-bearing components, has been meticulously designed to absorb the impact without significant deformation. Five seconds after deployment, the capture flaps close, drawing the net and the captured drone inside. The interceptor then utilises GPS navigation to fly to a predesignated safe zone, where it lands using a parachute, allowing the captured drone to be analysed and safely disposed of.

This detailed, technically sound interceptor concept was derived using a design strategy which identified the best existing propulsion and targeting technologies for this use case. This allowed for a focused, first principals-based investigation of the various aspects of the innovative capture mechanism. Consequently, this approach effectively simplified the complexity of the problem at hand, which was

essential given the limited timescale of the project. Specifically, it increased the attention paid to the various aspects of the capture mechanism, as exemplified in the highly detailed CAD of the bevel gears and the FEA analysis of the load-bearing elements. All in all, the work conducted throughout this project validated an innovative conceptual anti drone technology and outlined a pathway for its practical implementation.



Figure 15 – Interceptor with a Deployed Net

6.2 Implications and Further Work

A key consideration throughout the design process was the economic and social sustainability of the final product. Specifically, by focusing on the cost of materials and simplifying the design of the deployment mechanism, the manufacturing price for the system was minimised. This ensures that the retail cost of the interceptor can remain low, making this life-saving technology both commercially viable and widely available. This is particularly important for facilitating wide-scale adoption, including by rural communities and law enforcement in developing countries, who may not be able to afford an expensive system. However, the environmental sustainability of the design could still be improved by exploring alternative manufacturing methods to CNC machining. Despite its ability to produce mechanically strong parts, CNC machining is highly wasteful; therefore, alternatives such as additive manufacturing should be considered.

Due to the theoretical nature of this work, it is expected that some changes to the final product will be necessary as the prototype is built and tested. Specifically, this includes validating the dimensional tolerances between the bevel gears during manufacturing, as well as assessing the load-bearing capacity of the structural components through real-life static load tests. Additional work is also required

to validate the interceptor system as a whole. This includes conducting Simulink analysis to ensure that the rocket is aerodynamically stable with its atypical payload and specific component dimensions. Additionally, actual flight tests and interceptions must be carried out to confirm that all mechanisms function together as expected and to calibrate the various control systems of the interceptor.

6.4 Conclusion

This project successfully achieved its goal of designing a highly effective anti-drone system optimised for use in a civilian environment. To accomplish this, proven technological concepts were selected from various rocket systems and combined with a newly conceived capture mechanism. The detailed design outlined in this report has established the groundwork for prototyping this mechanism, as well as its integration with the targeting and propulsion systems. If successfully implemented on a large scale, this interceptor could significantly reduce the threat of drone-enabled terrorist attacks and other forms of misuse. Moreover, this threat, along with the risk of espionage, has led to some proposals for outright drone bans [29]. By providing authorities with an effective, safe, and accessible system to defeat drones, safe and highly desirable applications, such as photography and delivery services, can be enabled. It is recommended that this work is carried forward, including through the manufacturing and testing of a prototype.

7 Declarations

7.1. Acknowledgements: I am deeply grateful to Professor Giles Thomas for his persistent help.

7.2. Use of AI

AI was used in an assistive role (category 2).

8. References

- [1] Enterprise drone unit shipments www.statista.com/statistics/1234569/worldwide-enterprise-drone-market-shipments/2023, [2024 Aug 9].
- [2] Weaver M. The Gatwick drone <https://www.theguardian.com/uk-news/2020/dec/01/the-mystery-of-the-gatwick-drone> 2020 [2024 Aug 9].
- [3] Drone Terrorist Attacks www.ausa.org/sites/default/files/publications/LWP-137-The-Role-of-Drones-in-Future-Terrorist-Attacks_0.pdf [2024 Aug 9].
- [4] Physical barrier generation. Sci. Direct. <https://www.sciencedirect.com/science/article/pii/S2666647223000015>. 2023, [2024 Aug 9].
- [5] DRC <https://dronecatcher.nl/>. 2022, [2024 Aug 9].